From Long Term Classroom Programs and Medium / Short Classroom Program 4 in Top 10, 10 in Top 20, 43 in Top 100, 75 in Top 200, 159 in Top 500 Ranks & 3542 total selections in IIT-JEE 2012

ALL INDIA TEST SERIES

FIITJEE JEE(Advanced)-2013

ANSWERS, HINTS & SOLUTIONS
FULL TEST –IV
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	C	A
2.	A	B	C
3.	D	B	A
4.	B	C	C
5.	A	D	C
6.	C	C	B
7.	A	A	B
8.	A	C	D
9.	D	C	C
10.	C	A	B
11.	A	C	B
12.	C	B	B
13.	A	B	D
14.	B	D	B
15.	A, B, C	A,C	A, B
16.	A, B, C	A,C	A, B, C, D
17.	A, D	B,D	A, B, C
18.	A, B, D	B,D	C, D
19.	A, C, D	A, B, D	A, C
20.	C, D	B, C, D	A, C

Physics

PART – I

SECTION – A

1. A is at rest instantaneously (for no slipping at A)

$V_B - V_A = \frac{1}{2} B \left(\frac{V_0}{R}\right) (2R)^2$

$V_C - V_A = \frac{1}{2} B \left(\frac{V_0}{R}\right) (\sqrt{3}\, R)^2$

$\Rightarrow V_B - V_C = \frac{1}{2} B \left(\frac{V_0}{R}\right) [R^2] = \frac{BV_0R}{2}$

2. force $= (P - P_0) A = Ma$

by $P_1V_1 = P_2V_2$

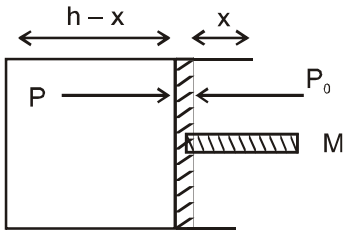
$P_0hA = P(h - x) A$

$\Rightarrow P = \frac{P_0h}{h - x}$

So force $= \left(\frac{P_0h}{h - x} - P_0\right) \frac{A}{M} = a$

$a = \frac{P_0Ax}{M(h - x)} \Rightarrow \frac{P_0Ax}{Mh}$; (for small x)

$T = 2\pi\sqrt{\frac{Mh}{P_0A}}$



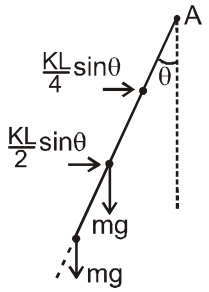
3. $\tau_{restoring} = (mg \sin\theta\, L + (mg \sin\theta\, \frac{L}{2}$

$+ \left(K\frac{L}{4}\sin\theta\right)\frac{L}{4} + \left(\frac{KL}{2}\sin\theta\right)\frac{L}{2}$

restoring $= \left[\frac{3mg}{2} + \frac{5KL^2}{16}\right]\theta$ (θ : small)

frequency of oscillation) $= \frac{1}{2\pi}\sqrt{\frac{\frac{3mg}{2} + \frac{5kL^2}{16}}{I}}$

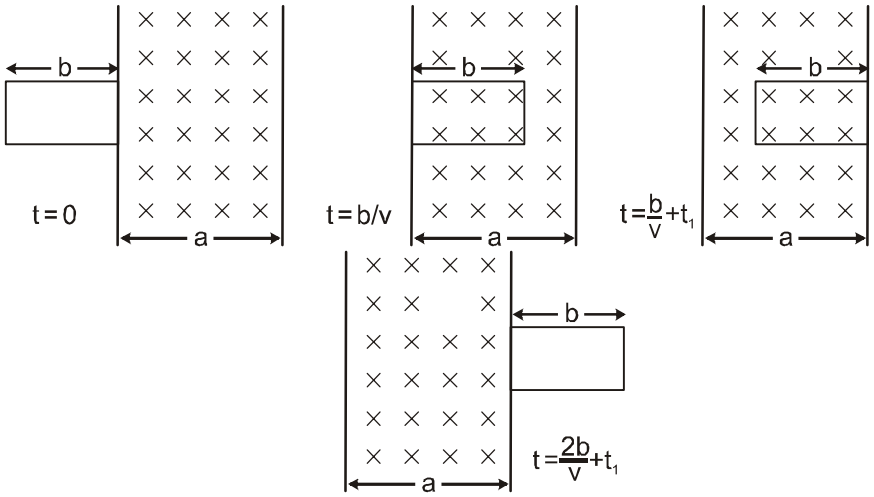
where $I = mL^2 + \frac{mL^2}{3} = \frac{4mL^2}{3}$



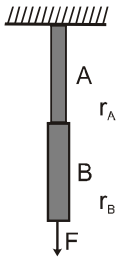
4. From law of radioactive decay.
Activity μ Number of nuclear of that nucleus present in the sample.
Activity does not depend upon the rate of the formation of nucleus.
In given reaction B decays into E and C.
Total activity of B = rate of decay of B into E + rate of decay of B into C = $l_5N_2 + l_3N_2$

5. The EMF will be induced in the loop, only when there is a change in the flux through the loop. When the loop was completely inside the field region, for some time (say t_1) there is no change in the flux, hence that time would not be included in the calculation.

The time for which EMF was induced = $\frac{b}{v}+\frac{b}{v}=\frac{2b}{v}$



6. That wire will break first whose stress reaches the breaking stress (for that material) first.



Stress = $\frac{F}{A}=\frac{F}{\pi r^2}$ where F is the tension in the wire. F is same in both the wires.

Thus stress $\propto \frac{1}{r^2}$ for the wire of smaller radius, the stress is more.

Thus the wire with smaller radius will break first.

If $r_A < r_B$, then wire A will break first.

If $r_A > r_B$, then wire B will break first.

If $r_A = r_B$, either of them could break.

8. As the magnetic field is uniform and the particle is projected in a direction perpendicular to the field, it will describe a circular path. The particle will not hit the yz-plane if the radius of the circle is smaller than d. For the maximum value of v, the radius is just equal to d. Thus $\frac{mv}{Bq}=d$.

$\therefore \quad v=\frac{Bqd}{m}$

9. $F = \eta A \frac{u-u_0}{y}$

$\Rightarrow u = \frac{F}{\eta A} y + u_0$

$\Rightarrow u = \frac{1}{10^{-2}\times 1} y + 1 = 100\, y + 1$

10. The velocity of upper plate is same as that of the layer in contact with the plate i.e. layer at y = 2 cm.

u = 100 (0.02) + 1 = 3 m/s

11. $T^2 \propto r^3$ (Kepler's law)

$$\frac{T_A^2}{T_B^2}=\frac{r_A^3}{r_B^3} = \frac{r^3}{(4r)^3}=\frac{1}{64} \text{ Get } \frac{T_A}{T_B}=\frac{1}{8}$$

12. For geostationary satellite, time period = 1 planet day (by def.)
Let T = 1 planet day
T_0 = 1 planet year

Now $T^2 = \frac{4\pi^2}{Gm}r_G^3 = \frac{4\pi^2}{Gm}r^3\left(\frac{m}{M}\right)$

$= \frac{4\pi^2}{GM}r^3 = T_0^3$ or $T = T_0$

13-14. $\omega = \sqrt{\frac{k}{M}}$, $\omega' = \sqrt{\frac{k}{M+m}} = \sqrt{\frac{2500}{25}} = 10$ rad/sec

$\omega' = \omega\sqrt{\frac{M}{M+m}}$

$x_1 = A \sin \phi_1$

Potential energy stored in the spring at ($t = t_1$) = $\frac{1}{2}kx_1^2 = \frac{1}{2}M\omega^2 \sin^2\phi_1$

Let v and v′ be the velocity of system just before collision and just after collision, so using COLM

$v' = \frac{Mv}{(M+m)} = \frac{MA\omega\cos\theta}{(M+m)}$

Total energy after collision = PE + KE = $\frac{1}{2}kx_1^2+\frac{1}{2}(M+m)v'^2$

$= \frac{1}{2}MA^2\omega^2\left[\frac{M+m\sin^2\theta}{M+m}\right] = \frac{1}{2}kA^2\left[\frac{M+m\sin^2\theta}{M+m}\right] = 8$ J

$\frac{1}{2}(M+m)A'^2\omega'^2 = \frac{1}{2}MA^2\omega^2\left[\frac{M+m\sin^2\theta}{M+m}\right]$

15. $\frac{I_{max}}{I_{min}}=\frac{(a_1+a_2)^2}{(a_1-a_2)^2}$

$\frac{a_1}{a_2}=\sqrt{\frac{I_1}{I_2}}=\sqrt{\frac{4}{9}}=\frac{2}{3}$

$\therefore \quad \frac{I_{max}}{I_{min}}=25$

If an identical paper is pasted across second slit shifted C.B.F. will back in central point.

$$\text{shift} = (\mu-1)\frac{tD}{d}$$

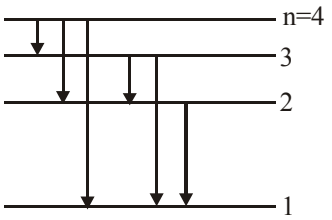
$$\text{fringe width} = \frac{n\lambda D}{d}$$

$\therefore \quad n = \frac{\text{shift}}{\text{fringe width}} = \frac{(\mu-1)t}{\lambda} = 15$.

16. The change in magnetic flux is zero, hence the current in the ring will be zero.

17. Both (A) and (D) depend on g.

18. $\frac{n(n-1)}{2} = 6 \Rightarrow n = 4$



If the initial state were n = 3, in the emission spectrum, no wavelengths shorter than λ_0 would have occurred. This is possible if initial state were n = 2

$\therefore$ (A), (B) and (D)

19. From COE (A) is correct.

The force outside the earth varies as inverse square of the distance.

$\therefore$ motion is not simple harmonic. However, from symmetry of motion, the motion will be periodic.

$\therefore$ (C) is correct.

From COE

$$\frac{1}{2}mv^2 = -\frac{GMm}{2R} - \left(-\frac{3GMm}{2R}\right)$$

$$\Rightarrow \quad v = \sqrt{\frac{2GM}{R}}$$

$\therefore$ (A), (C) and (D) are correct.

20. The current through R_1 is constant.

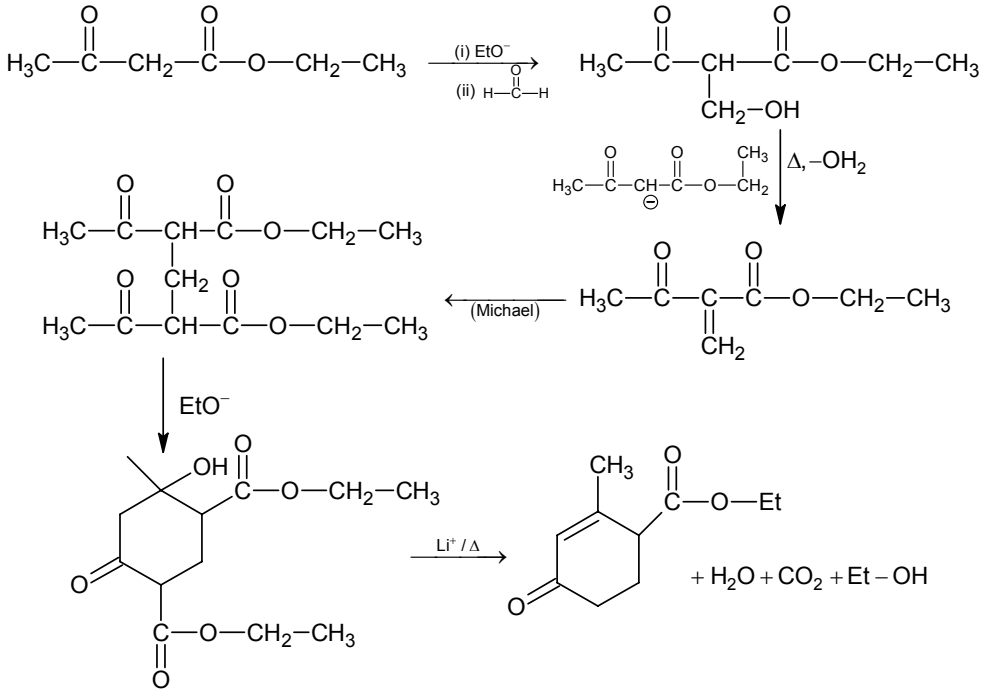
Chemistry PART – II

SECTION – A

1. pH at isoelectric point = $\frac{pKa_1 + pKa_2}{2} = \frac{4+9}{2} = 6.5$

2. Net reaction does not involve any aqueous species.

5. $\left[Cr(SCN)_2(NH_3)_4\right]^{2+}$, shows geometrical (or cis-trans) and linkage isomerism.

6. 

7. $FeSO_4 \xrightarrow{\Delta} Fe_2O_3 + SO_3 + SO_2$

8. Energy of e^- in H-atom = - E
Energy of e^- in Li^{+2} = - 9E
Energy supplied by photon = $IE + \frac{1}{2}mv^2$
$IE = |E|$
$\therefore E_p = 9E + \frac{1}{2}mv^2$
$v = \sqrt{\frac{2(E_p - 9E)}{m}}$

9. In reverse osmosis water moves from more concentrated to less concentrated solution

10. π = iCRT, (i=2)

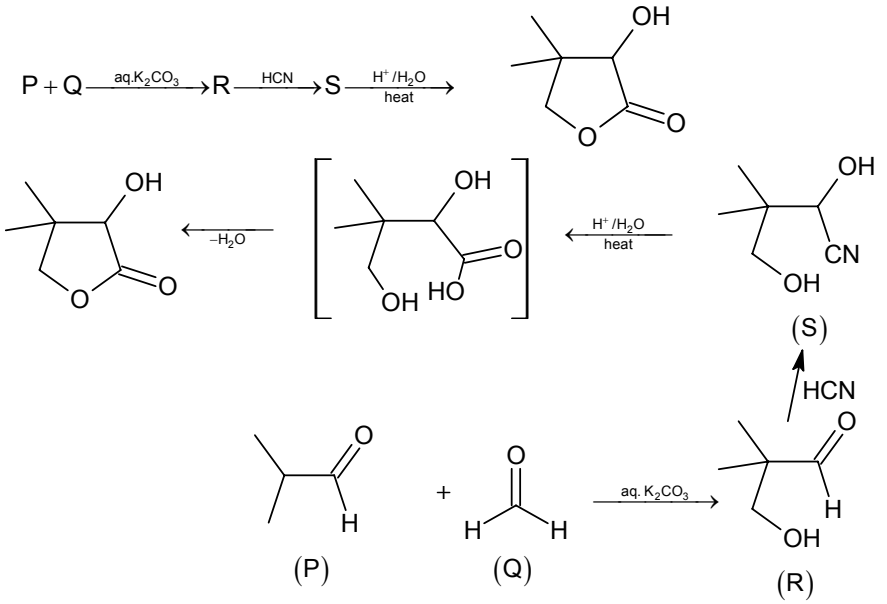
11. Let conc. of both reactants at equilibrium is x M

$$Keq = \frac{\left[PbEDTA^{2-}\right]\left[H^{+}\right]^{2}}{x^{2}}$$

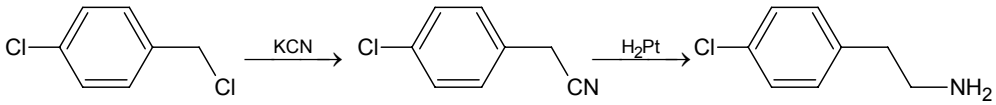
$x = 10^{-3}$

12. Chelation increases stability of complexes

14.



15.

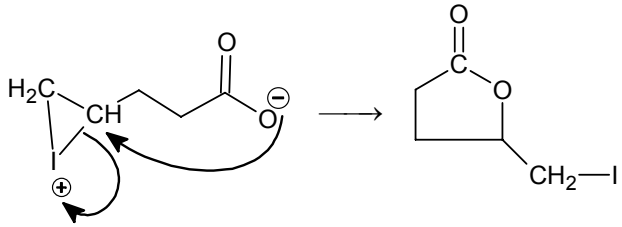


16. Trans-2-butene $\xrightarrow{CF_3CO_3H}$ epoxide $\xrightarrow{H_3O^+}$ diol $\xrightarrow{HIO_4}$ $2CH_3CHO$

17. He and H_2 are not hydrogen like particles

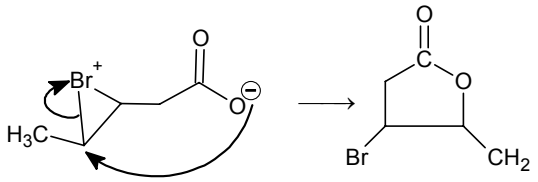
18. Radius of cation is smaller than parent atom and that of anion is more than parent atom

19. (A)



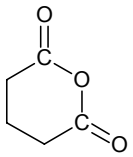
I_2 will attack the double bond to form a cyclic halonium ion. It then undergoes on intramolecular reaction. Formation of five membered ring is favoured because (i) attack by a nucleophile on a halonium ion normally takes place at the more substituted site.
(ii) Formation of five membered ring is more probable.

(B)



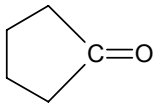
(C)

Option(C)gives



(D)

Option(D)gives



1, 4 and 1,5-dicarboxylic acids give anhydrides whereas 1,6 and 1, 7-diacids give ketones when heated with P_2O_5.

Mathematics

PART – III

SECTION – A

1. Normal to $y^2 = 4ax$ with slope m is

$y = mx - 2am - am^3$ ..... (1)

(1) touches $x^2 - y^2 = a^2$ if

$(-2am - am^3)^2 = a^2 (m^2 - 1)$

i. e. if $m^6 + 4m^4 + 3m^2 + 1 = 0$ ..... (2)

(2) is a sixth degree equation in m with no odd power terms and whose coefficients are all positive.

$\Rightarrow$ (2) has no real root

$\therefore |S| = 0$

2. Given $x + y + z = 5 \rightarrow (1)$

and $x^2 + y^2 + z^2 = 9 \rightarrow (2)$

We have $yz = \frac{1}{2}\left[(y+z)^2 - \left(y^2 + z^2\right)\right]$

$= \frac{1}{2}\left[(5-x)^2 - \left(9 - x^2\right)\right]$ (using) (1) & (2))

$= x^2 - 5x + 8$

$\Rightarrow$ y, z must be the roots of the equation

$u^2 - (5-x)u + \left(x^2 - 5x + 8\right) = 0 \rightarrow (3)$

As y, z are real, we must have discriminant of (3) ≥ 0

$\Rightarrow (5 - x)^2 - 4(x^2 - 5x + 8) \geq 0$

$\Rightarrow 3x^2 - 10x + 7 \leq 0$

$\Rightarrow (3x - 7)(x - 1) \leq 0$

$\Rightarrow x \in \left[1, \frac{7}{3}\right]$

$\Rightarrow$ Number of integer values that x can take is 2

3. $2\sin^2 x + \sin^2 2x = 2$

$\Rightarrow -2(1 - \sin^2 x) + 4\sin^2 x \cos^2 x = 0$

$\Rightarrow \cos^2 x . \cos 2x = 0 \rightarrow (1)$

$\sin 2x + \cos 2x = \tan x$

$\Rightarrow 2\sin x \cos^2 x + \cos 2x \cos x = \sin x$

$\Rightarrow \sin x \cos 2x + \cos 2x \cos x = 0$

$\Rightarrow \cos 2x (\sin x + \cos x) = 0 \rightarrow (2)$

Common roots of (1) and (2) are given by $\cos 2x = 0$

$\Rightarrow x = \frac{(2n-1)\pi}{4}, n \in N$

In $[0, 4\pi]$ we have $x = \frac{n\pi}{4}$, where n = 1, 3, 5,……,15 $\rightarrow (3)$

Now, $-2\sin^2 x + \sin x \leq 0$

$\Rightarrow 2\sin^2 x - \sin x \geq 0$

$\Rightarrow (\sin x)(2\sin x - 1) \geq 0$

$\Rightarrow \sin x \in [-1, 0]$ or $\frac{1}{2} \leq \sin x \leq 1 \rightarrow (4)$

Among the 8 values given by (3), only 4 values, i.e., $x = \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{13\pi}{4}, \frac{15\pi}{4}$ satisfy (4)

4. We have $w = z + \frac{1}{z}$

Take $z = e^{i\theta}$ ($\because |z| = 1$).

We have $w = e^{i\theta} + e^{-i\theta} = 2\cos\theta$

We know that w is real and $|w| \le 2$

5. The scalar triple product of the three given vectors is $\begin{vmatrix} x^2-1 & 2(x^2-1) & -3(x^2-1) \\ 2x^2-1 & 2x^2+1 & x^2 \\ 3x^2+2 & x^2+4 & x^2+1 \end{vmatrix}$

$$= (x^2-1)\begin{vmatrix} 1 & 2 & -3 \\ 2x^2-1 & 2x^2+1 & x^2 \\ 3x^2+2 & x^2+4 & x^2+1 \end{vmatrix} = (x^2-1)\begin{vmatrix} 1 & 2 & 0 \\ 2x^2-1 & 2x^2+1 & 5x^2 \\ 3x^2+2 & x^2+4 & 5x^2+7 \end{vmatrix} \text{ (using } C_3 \to C_3 + C_2 + C_1)$$

$$= (x^2-1)\begin{vmatrix} 1 & 0 & 0 \\ 2x^2-1 & -2x^2+3 & 5x^2 \\ 3x^2+2 & -5x^2 & 5x^2+7 \end{vmatrix} \text{ (using } C_2 \to C_2 - 2C_1)$$

$= (x^2-1)(15x^4 + x^2 + 21)$

Vector are non–coplanar $\Rightarrow$ Scalar triple product $\ne 0 \Rightarrow x^2 - 1 \ne 0$ ($\because 15x^4 + x^2 + 21 \ne 0$) $\Rightarrow x \ne \pm 1$

6. We have $f(x) = (a - x^n)^{1/n}$ so that

$$f \circ f(x) = f\left((a-x^n)^{1/n}\right) = \left(a - \left((a-x^n)^{1/n}\right)^n\right)^{1/n} = \left(a - (a - x^n)\right)^{1/n}$$

$$\Rightarrow g(x) = \underbrace{f \circ f \circ \ldots\ldots f(x)}_{2m \text{ times}} = x$$

$$\therefore \int \frac{dx}{(g(x))^n \left(1 + (g(x))^n\right)^{1/n}} = \int \frac{dx}{x^n (1+x^n)^{1/n}} = \int \frac{dx}{x^{n+1}(1+x^{-n})^{1/n}} = -\int \frac{dt}{nt^{1/n}} \text{ (using } t = 1 + x^{-n})$$

$$= -\frac{1}{n}\left(\frac{t^{-1/n+1}}{-\frac{1}{n}+1}\right) + c = \frac{1}{1-n}\left(1 + x^{-n}\right)^{1-\frac{1}{n}} + c$$

7. Given $y = f(x), z = g(x)$ we have

$$\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{f'(x)}{g'(x)} \text{ and } \frac{d^2y}{dz^2} = \frac{d}{dx}\left(\frac{f'}{g'}\right)\frac{dx}{dz} = \frac{g'f'' - f'g''}{(g')^2} \cdot \frac{1}{g'}$$

8. $f(x)$ is continuous at x = 3

$\Rightarrow f(3^-) = f(3) = f(3^+)$

Now $f(3^-) = \lim\limits_{x \to 3^-} \frac{a|x^2 - x - 6|}{6 + x - x^2} = \lim\limits_{x \to 3^-} \frac{-a(x^2 - x - 6)}{6 + x - x^2} = a$

$$f\left(3^+\right) = \lim_{x\to 3^+} \frac{x-[x]}{x-3} = \lim_{x\to 3^+} \frac{x-3}{x-3} = 1$$

$$f(3) = b \lim_{x\to 3}\left(\frac{x\int_3^x tdt}{x-3}\right)\left(\frac{0}{0}\text{form}\right) = b \lim_{x\to 3}\left(\frac{x^2+\int_3^x tdt}{1}\right) = 9b$$

$\Rightarrow a = 9b = 1 \Rightarrow b = \frac{1}{9}$ and $a = 1$

9. $V_n = \sum_{r=1}^{n} S_r = 2\sum_{r=1}^{n} r = n(n+1)$

$\therefore \lim_{n\to\infty} \frac{V_n}{n^2} = \lim_{n\to\infty} \frac{n+1}{n} = 1$

10. (i) $V_n = n(n+1) \Rightarrow V_1 < V_2 < V_3 < .....$

$\Rightarrow$ Statement – 2 is true

(ii) $Q_n = \sum_{r=1}^{n} \frac{1}{S_r S_{r+1}} = \frac{1}{4}\sum_{r=1}^{n} \frac{1}{r(r+1)} = \frac{1}{4}\sum_{r=1}^{n}\left(\frac{1}{r} - \frac{1}{r+1}\right) = \frac{1}{4}\left(1 - \frac{1}{n+1}\right)$

$\Rightarrow Q_1 < Q_2 < Q_3 < \ldots\ldots\ldots$

$\Rightarrow$ Statement –1 is false

11. We have

$$f(x) = \begin{cases} 2a^{-x}, & x < 0 \\ a^{-x} + a^{x}, & x \geq 0 \end{cases}$$

Clearly, f(x) is continuous in R.

For $x > 0$, $f'(x) = -a^{-x}\log_e a + a^x \log_e a = (a^x - a^{-x})\log_e a = \left(\frac{a^{2x}-1}{a^x}\right)\log_e a > 0$ for $x > 0$

$\Rightarrow f(x)$ is monotonic increasing for $x > 0$

For $x < 0$, $f(x) = 2a^{-x} \Rightarrow f'(x) = -2a^{-x}\log_e a < 0$

$\Rightarrow f(x)$ is monotonic decreasing for $x < 0$

12. Required area $= \int_{-1}^{0} 2a^{-x}dx + \int_{0}^{1}\left(a^{-x} + a^{x}\right)dx = \left\{\frac{-2a^{-x}}{\log_e a}\right\}_{-1}^{0} + \left\{\frac{-a^{-x}+a^x}{\log_e a}\right\}_{0}^{1}$

$$= \frac{1}{\log_e a}\left[-2-(-2a)\right] + \frac{1}{\log_e a}\left[\left(\frac{-1}{a}+a\right)-0\right] = \frac{3a^2-2a-1}{a\log_e a}$$

15. The equation of the line is $b^{\frac{1}{3}}y + a^{\frac{1}{3}}x + a^{\frac{2}{3}}b^{\frac{2}{3}} = 0$

This can be written as $y = -\frac{a^{\frac{1}{3}}}{b^{\frac{1}{3}}}x - a^{\frac{2}{3}}b^{\frac{1}{3}} \Rightarrow m = \frac{-a^{\frac{1}{3}}}{b^{\frac{1}{3}}}$, $c = -a^{\frac{2}{3}}b^{\frac{1}{3}}$

Consider the parabola $y^2 = 4ax$

$$\frac{a}{m} = \frac{a}{\frac{-a^{\frac{1}{3}}}{b^{\frac{1}{3}}}} = -\frac{ab^{\frac{1}{3}}}{a^{\frac{1}{3}}} = -a^{\frac{2}{3}}b^{\frac{1}{3}} = c$$

$\therefore$ Condition for tangency for the parabola $y^2 = 4ax$ is satisfied

$\therefore$ The given line touches $y^2 = 4ax$

Similarly the line touches $x^2 = 4by$

16. $(x+iy)^{\frac{1}{5}} = a+ib$

$\therefore x + iy = (a + ib)^5 = a^5 + 5a^4 (ib) + 10a^3 (-b^2) + 10a^2 (-ib^3) + 5ab^4 + ib^5$

Equating the real and imaginary parts $x = a^5 - 10a^3b^2 + 5ab^4 \Rightarrow y = 5a^4b - 10a^2b^3 + b^5$

$p = \frac{x}{a} - \frac{y}{b} = (a^4 - 10a^2b^2 + 5b^4) - (5a^4 - 10a^2b^2 + b^4) = -4a^4 + 4b^4 = -4(a^4 - b^4)$

$= -4(a^2 - b^2)(a^2 + b^2) = -4(a + b)(a - b)(a + ib)(a - ib)$

$\therefore$ a – b, a + b, a + ib and a – ib are all factors of p.

17. Since $0 < \theta < \frac{\pi}{4}, 0 < \sin^2\theta, \cos^2\theta, \tan^2\theta < 1$

$$(A)\sum_{k=0}^{\infty}\sin^{2k}\theta = \frac{1}{1-\sin^2\theta} = \sec^2\theta$$

$$(B)\ \sum_{k=0}^{\infty}\cos^{2k}\theta = \frac{1}{1-\cos^2\theta} = \operatorname{cosec}^2\theta$$

$$(C)\ \sum_{k=0}^{\infty}\tan^{2k}\theta = \frac{1}{1-\tan^2\theta} = \frac{\cos^2\theta}{\cos^2\theta - \sin^2\theta} = \frac{\cos^2\theta}{\cos 2\theta}$$

(D) $\cot\theta > 1$ in $0 < \theta < \frac{\pi}{4}$. Hence, $\sum_{k=0}^{\infty}\cot^{2k}\theta$ does not exist.

19. $3^y = 9 - 3^x \Rightarrow 9 - 3^x > 0 \Rightarrow x < 2$

Similarly, $y < 2$

20. $\text{Area}\ (T) = \frac{c.c^2}{2} = \frac{c^3}{2}$

$$\text{Area}\ (R) = \frac{c^3}{2} - \int_0^c x^2 dx$$

$$= \frac{c^3}{2} - \frac{c^3}{3} = \frac{c^3}{6}$$

$$\lim_{c\to 0^+}\frac{\text{Area}\ (T)}{\text{Area}\ (R)} = \lim_{R\to 0^+}\frac{c^3}{2}.\frac{6}{c^3} = 3$$